

10.2: VSEPR

Introduction:

This part of the review goes over the use of Valence Shell Electron Repulsion Theory (VSEPR) to determine molecular geometries. You should review your general chemistry 1 notes. If they are not available, go to [section 8.6: Molecular Geometries](#)

Worksheets

Handout 10.2.1 : Overview of Structures:

http://chemwiki.ucdavis.edu/@api/deki/files/60818/VSEPR_Geometry_Handout.pdf

The above 3 page handout gives the basic VSEPR geometries of table 2.1 (below), along with the hybrid orbitals (review section)

Handout 10.2.2 : VSEPR Worksheet

http://chemwiki.ucdavis.edu/@api/deki/files/61155/vsepr2_Worksheet.pdf

The above handout is a continuation of the exercise in section 10.1 Lewis Dot Structures. You should fill out that handout first, and then use it while filling out this handout.

Molecular Geometries

Table 1 shows some examples of geometries with a central atom A is bonded to two or more X atoms. As indicated in several of the geometries below, non-bonding electrons can strongly influence the molecular geometry of the molecule. Note: "E" represent lone pairs of electrons

Table 10.2.1 :

6	5	4	3	2
AX_6  octahedral	AX_5  trigonal bipyramidal	AX_4  tetrahedral	AX_3  trigonal planar	AX_2 Linear linear
1 lone pair of electrons				
AX_5E  square pyramidal	AX_4E  distorted tetrahedron	AX_3E  pyramidal	AX_2E  nonlinear	AXE Linear linear
2 lone pairs of electrons				
AX_4E_2  square planar	AX_3E_2  T-shaped	AX_2E_2  bent		

These structures can generally be predicted, when A is a nonmetal, using the "valence-shell electron-pair repulsion model (VSEPR) discussed in the next section. This image table was borrowed from Paul Groves LibreText, and the rotating molecules were created by Robyn Rindge.

Exercise 10.2.1

What is the molecular geometry of triiodide (I_3^-), which has two bonding and 3 lone pairs? (It is not in the above table)

Answer

Linear

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